

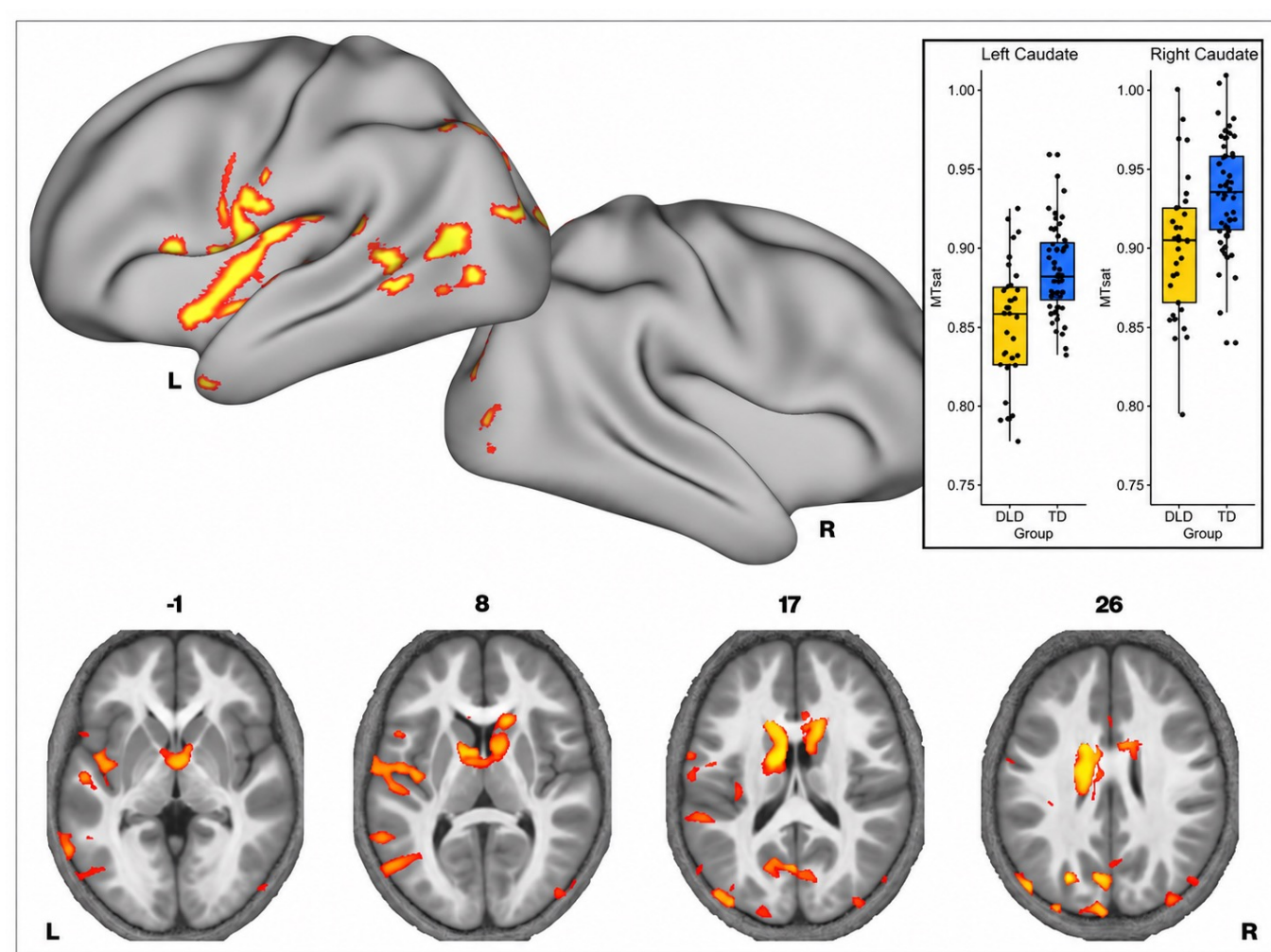
Longitudinal changes in caudate nucleus myelination in developmental language disorder

Nilgoun Bahar^{1,2}, Saloni Krishnan³, Kate E. Watkins¹

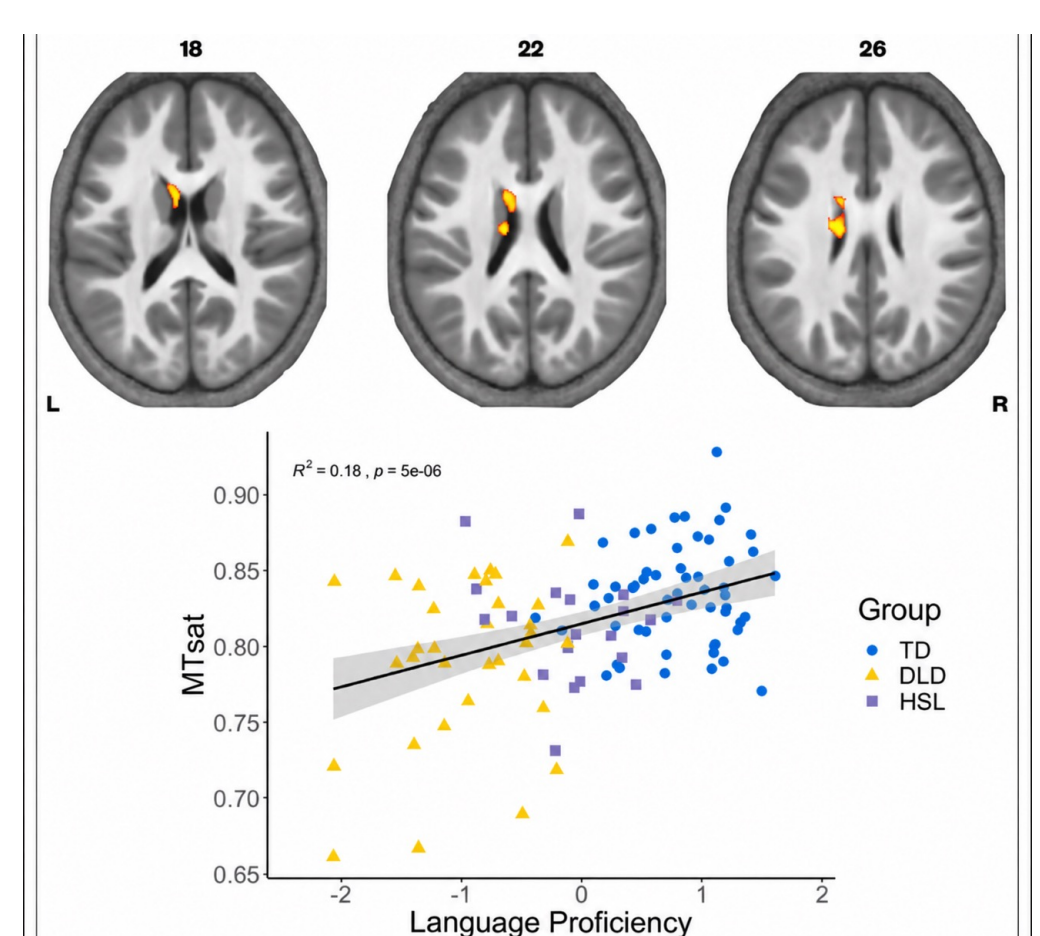
¹ Department of Experimental Psychology, University of Oxford; ² Department of Neurology, Dyslexia Center, UCSF; ³ Division of Psychology & Language Sciences, University College London, UK

Background & Research Questions

- Developmental language disorder (DLD) affects ~7% of children and is characterised by persistent difficulties in learning and using language.
- Theoretical accounts implicate **frontostriatal circuitry**, particularly the **dorsal striatum**, which supports procedural learning skills important for language acquisition.
- Using quantitative MRI, **Krishnan et al. (2022)** found lower myelin-sensitive MTsat values in the caudate nucleus in 10-to-15-year-olds with DLD, suggesting striatal microstructural differences.

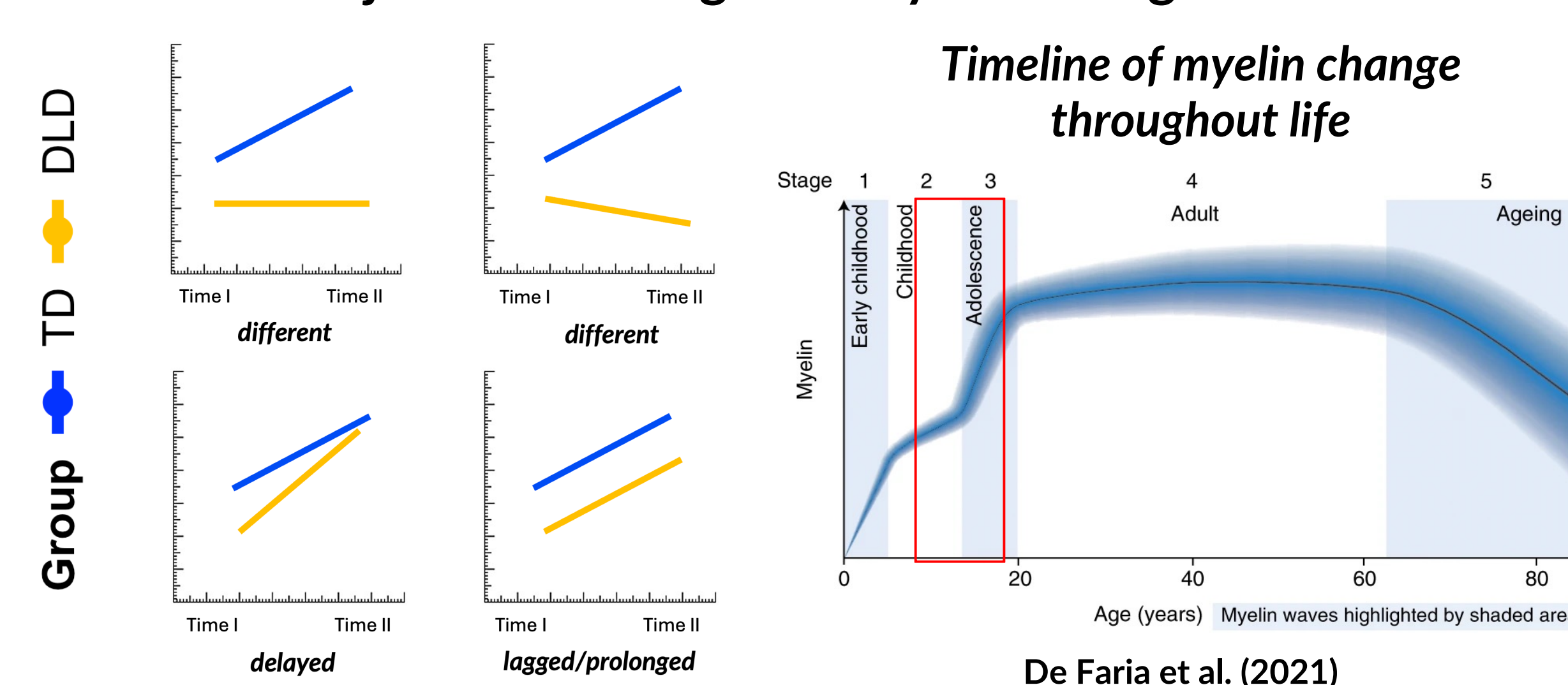


- MTsat in caudate nucleus was positively correlated with language proficiency across the cohort (both TD and DLD).
- Developmental course of microstructural myelin differences is unknown in DLD.



Q: Do caudate nucleus microstructural differences in DLD persist across adolescence, or do they reflect delayed maturation that normalises over time?

- Possible trajectories of regional myelin change in DLD:

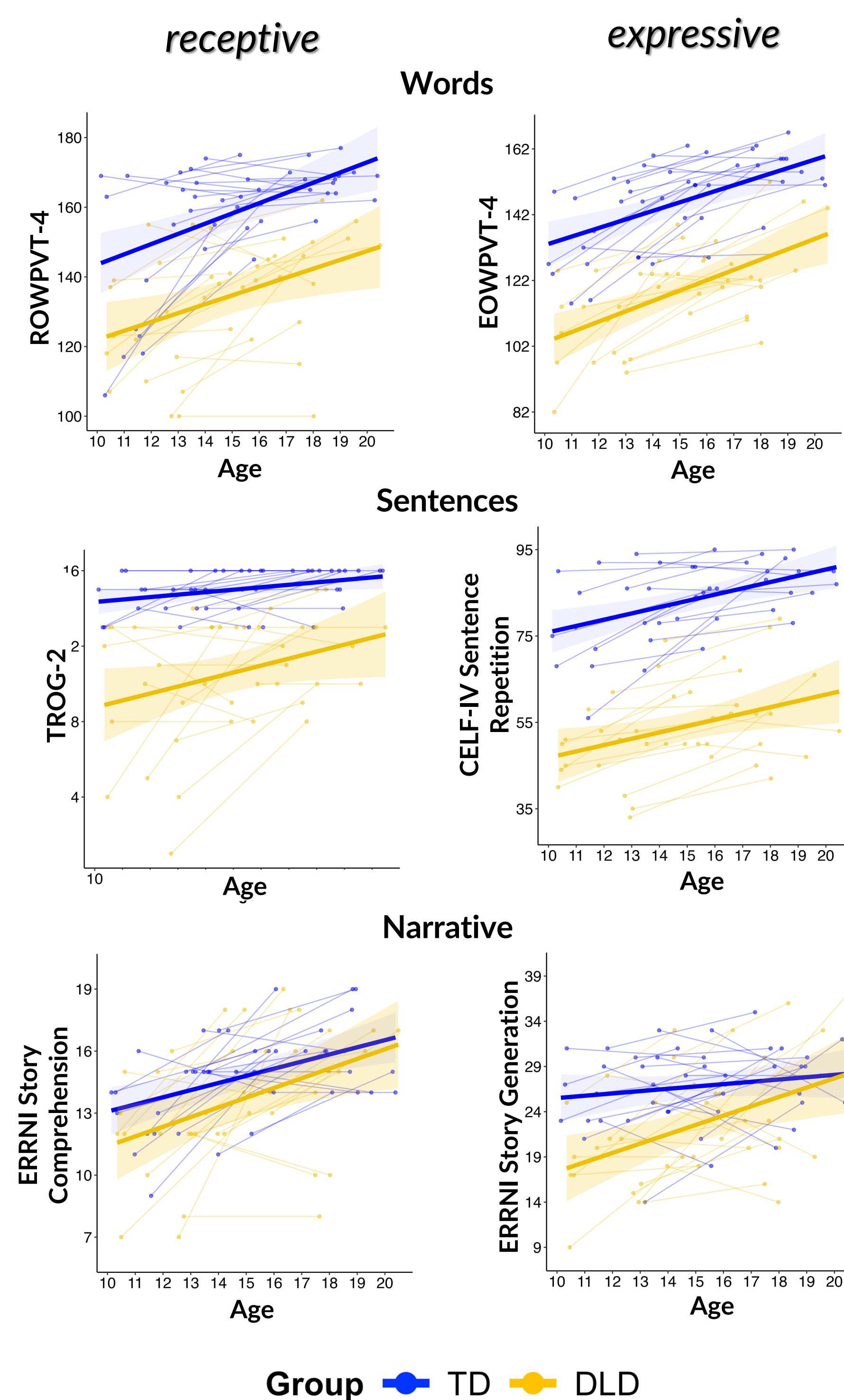


Participants & Group Definitions

- N = 41 Native English-speaking school-age children
- Time I age: 10-15 years ($m = 12.61$ y)
- Time II age: 14-20 years ($m = 17.22$ y)
- Mean interval between sessions = 4y6m (3y9m – 5y11m)

	Nonverbal reasoning >70	2+ language tests failed	History of speech & language difficulties
22 TD 11 Male 18 R-handed	✓	X	X
19 DLD 12 Male 14 R-handed	✓	✓	✓

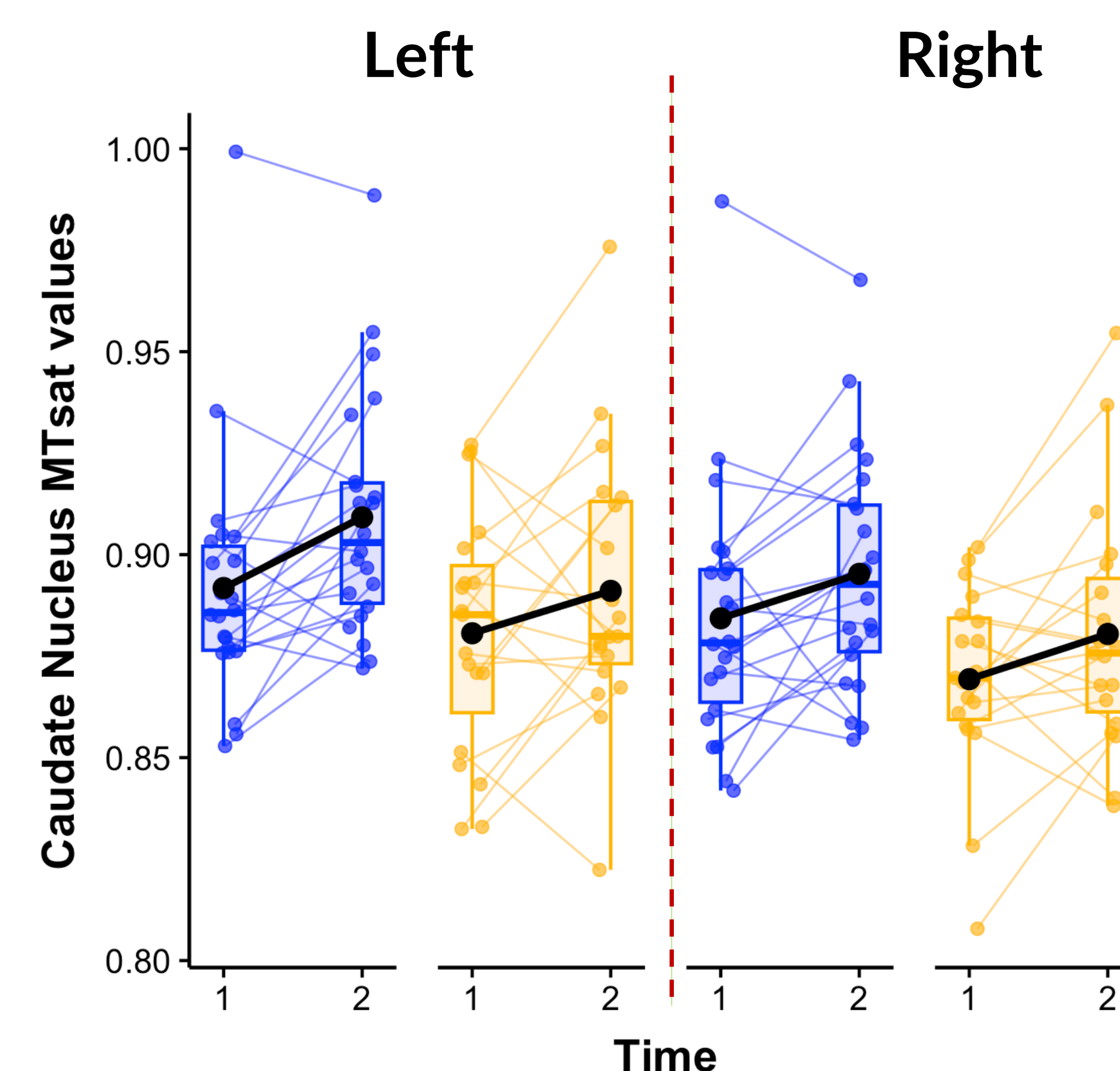
Changes in language scores over time



- Similar rates of progress in TD and DLD in language scores over time, but a consistent performance gap remained in DLD in tests of vocabulary and grammar, especially on sentence repetition test (proxy for expressive grammar).

Results

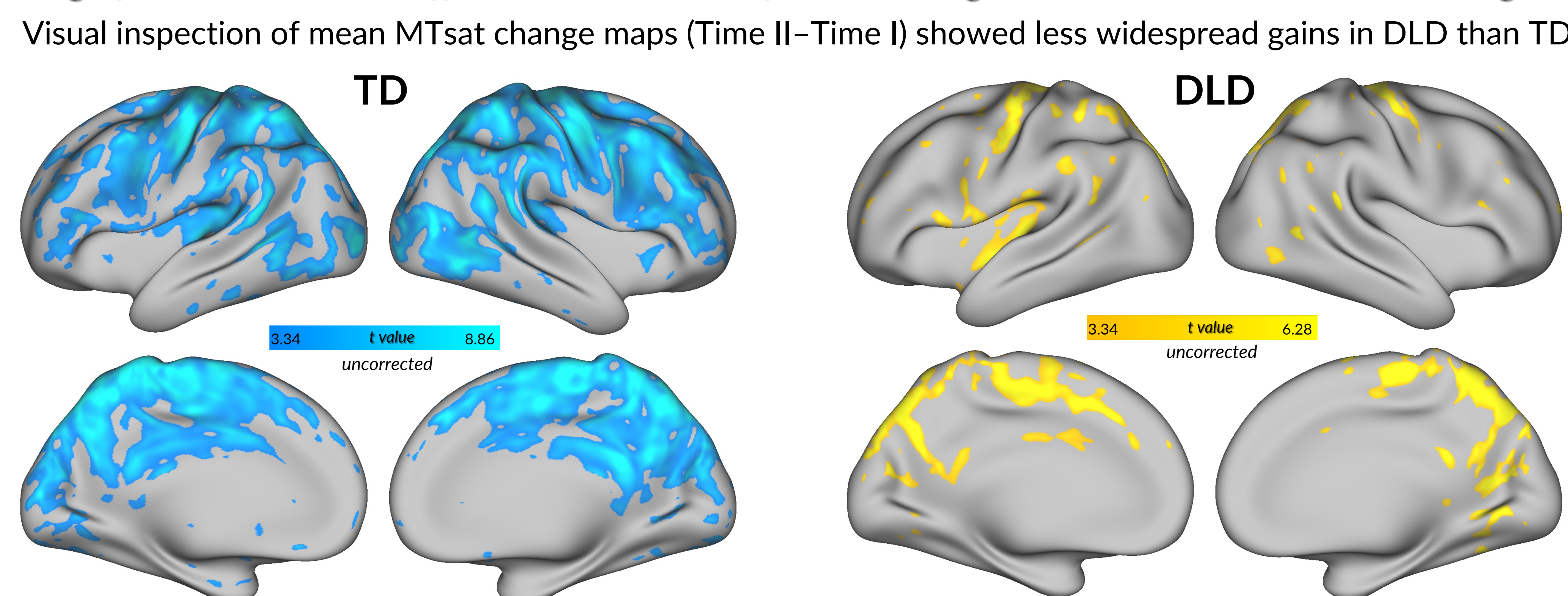
Changes in MTsat values (myelin index) over time (ROI + whole brain)



Group ● TD ● DLD

- Lower caudate nucleus MTsat values in DLD than TD across time and hemisphere, $F(1, 39) = 3.34$, $p = .035$ (planned one-tailed comparison)
- MTsat increased significantly over time across groups and hemispheres, $F(1, 39) = 11.41$, $p = .001$
- No evidence of different rate of change in caudate nucleus MTsat values between DLD and TD (Group $\times$ Time, $p = .653$)

No significant whole brain differences in the rate of MTsat change over time between TD and DLD groups



Materials & Methods

Neuropsychological Measures

- Expressive/receptive vocabulary, grammar, and narrative skills

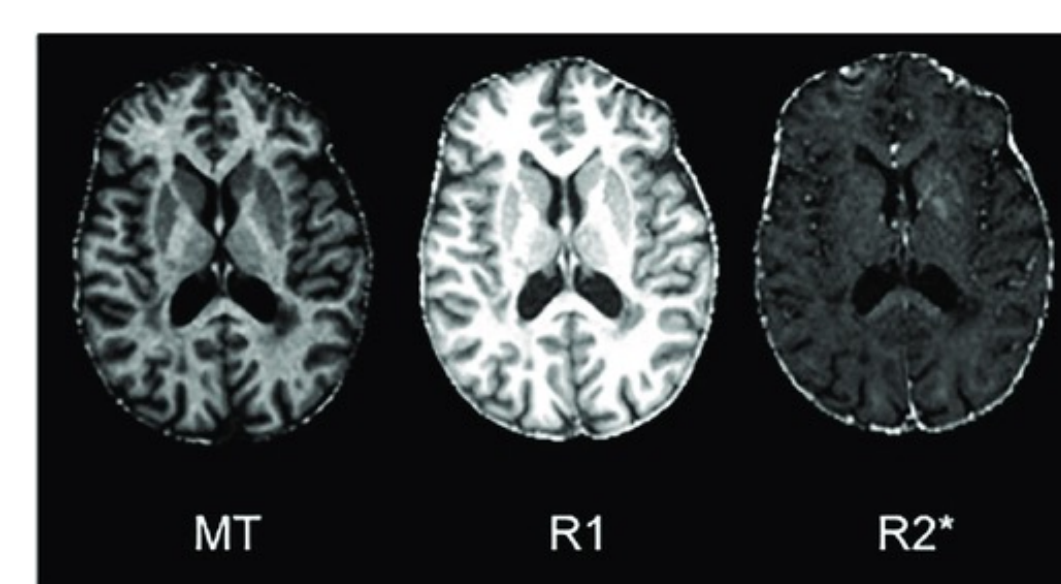
Statistical Approach

- Behav Outcome = $\beta_0 + \beta_1(\text{Group}) + \beta_2(\text{Time}) + \beta_3(\text{Group} \times \text{Time}) + \epsilon$

Imaging Measures

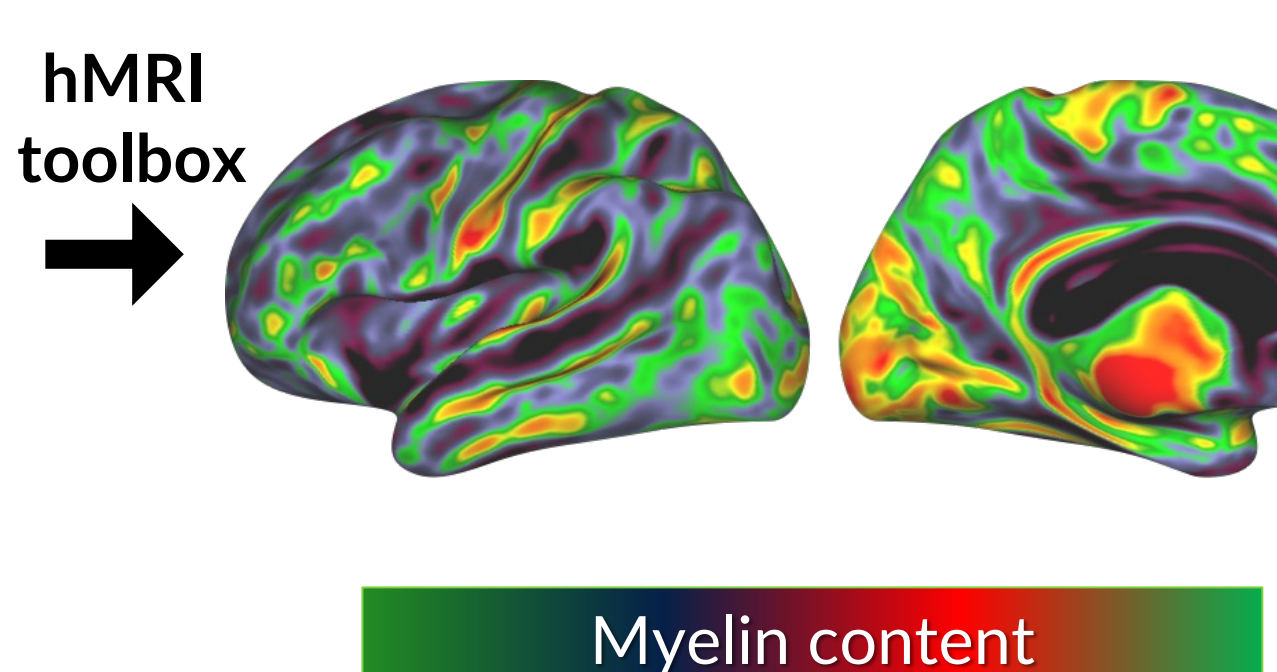
Histological MRI: Quantitative multiparameter maps

Acquired images



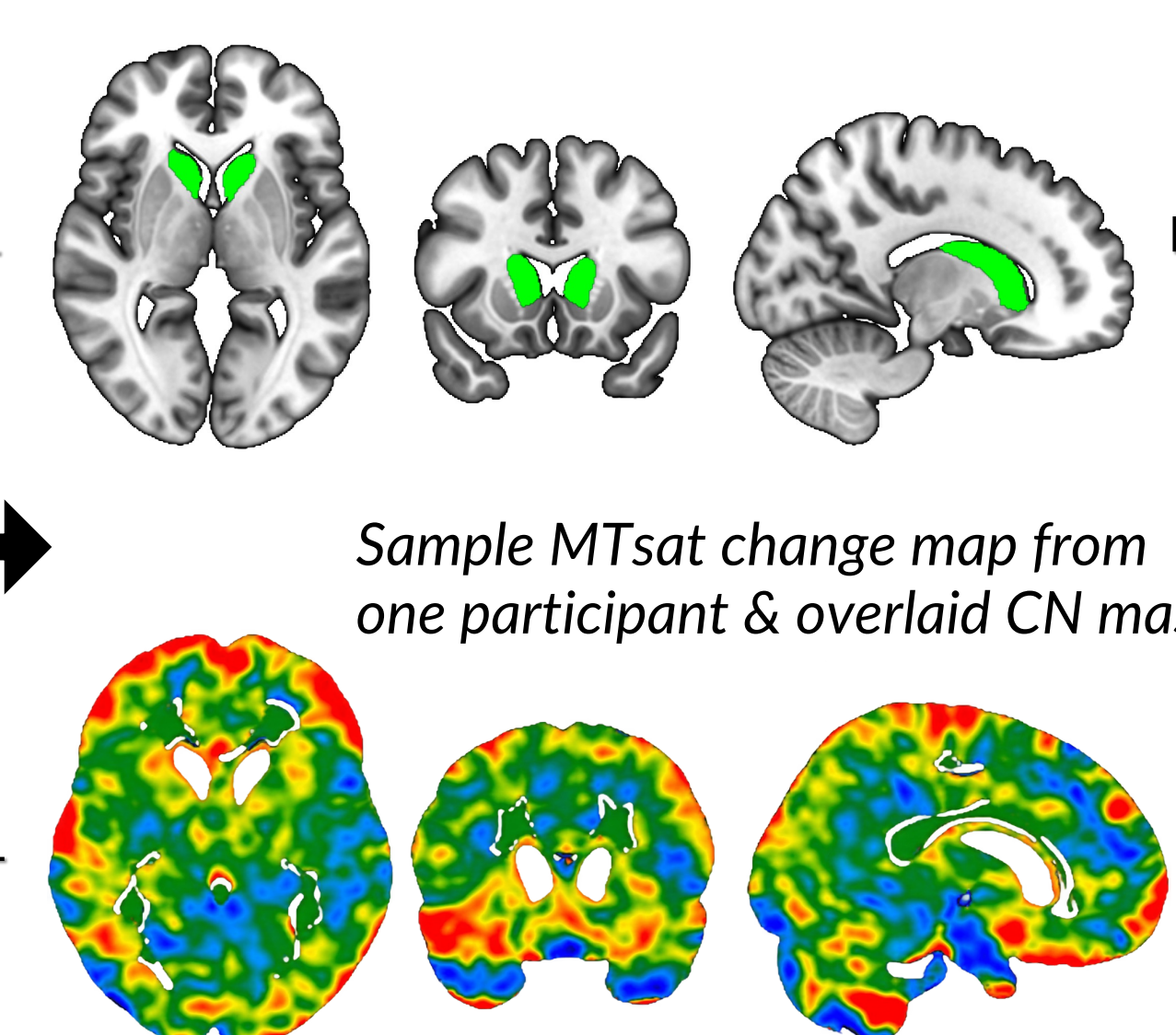
Detailed & reproducible assessment of neural architecture

Estimated magnetization saturation (MTsat) maps



(Weiskopf et al., 2013 | *Frontiers in Neuroscience*)

MTsat change maps (Time II–Time I) + caudate nucleus mask



Conclusions

- Behaviour:** Qualitatively similar developmental patterns in DLD and TD groups, but DLD group shows consistently lower scores across most language skills, reflecting a persistent performance lag extending into adolescence and young adulthood.
- Brain:** Participants with DLD showed persistently lower caudate nucleus MTsat values across timepoints, with no evidence of catch-up. This stable group difference suggests that altered caudate nucleus microstructure is a persistent neurodevelopmental feature of DLD rather than a temporary maturational delay.